

BilgeGuard — No-Subscription Boat Monitor

Comprehensive Business Plan & Market Analysis

Confidential — Draft v1.0 / 2026

EXECUTIVE SUMMARY

Company: BilgeGuard (working name) **Product:** A bring-your-own-SIM marine monitoring device (bilge, battery, GPS) with no mandatory subscription **Launch vehicle:** Kickstarter crowdfunding campaign **Kickstarter goal:** \$50,000 (conservative; engineered for early overfunding momentum) **Unit price:** \$199 **BOM cost:** ~\$35 **Gross margin:** ~82% **Target market:** ~11.8 million registered US recreational boats **Total addressable market (US):** ~\$2.35 billion (at 1% penetration of registered fleet at \$199)

1. THE PROBLEM

There are currently 11.8 million registered recreational boats in the United States (NMMA, 2024). A significant portion of these boats sit unattended at marinas, on trailers, or at anchor for days or weeks at a time. Three acute risks exist for every unattended boat owner:

1. **Silent sinking** — bilge pump failure is the leading non-weather cause of boats sinking at the dock. Owners often don't know until they arrive to find their boat gone.
2. **Battery death** — a dead battery from parasitic draw or a failed charge cycle means a boat that won't start, expensive tow bills, and a ruined day on the water.
3. **Theft** — outboard motors (commonly \$10,000–\$30,000), electronics, and entire boats are stolen regularly. Without GPS monitoring, recovery is rare.

The current "solution" is broken. Existing products such as Siren Marine (acquired by Yamaha) and BRNKL charge \$299–\$799 upfront AND lock users into mandatory cellular subscriptions of \$15–\$25/month. A Siren 3 Pro at \$799 + \$18/month = **\$1,015 in year one alone** to protect a boat. This is widely considered the most-hated subscription in recreational boating forums (The Hull Truth, Trawler Forum, Reddit r/boating).

The market signal is clear: DIY threads asking "how do I build my own boat monitor" appear regularly on every major boating forum, confirming that demand exists but the price and subscription model are blocking adoption.

2. THE SOLUTION — BILGEGUARD

BilgeGuard is a weatherproof ESP32-based monitoring device that:

- Monitors **bilge float switch** (sinking alert within seconds)
- Monitors **battery voltage** (configurable low-voltage alert thresholds)
- Tracks **GPS location** (geofence alerts, theft detection)
- Sends alerts via **SMS and push notification** using the owner's own IoT SIM card (\$2–

\$4/month from Hologram, Twilio, or T-Mobile IoT) **OR** connects to marina WiFi at zero cellular cost

- Uses **open MQTT protocol** — no proprietary cloud lock-in
- Is **user-installable** in 30-45 minutes (two wires to battery, one float switch clip, done)

What it is NOT: It is not another subscription trap. The owner buys the hardware once. The cellular costs go directly to a commodity SIM provider at commodity rates.

Core Hardware Stack

Component	Purpose	Unit Cost
ESP32-S3 microcontroller	Main brain	~\$4
SIM7080G LTE-M/NB-IoT module	Cellular connectivity	~\$12
u-blox MAX-M10S GPS	Location tracking	~\$6
Float switch input circuit	Bilge detection	~\$2
Voltage divider + ADC	Battery monitoring	~\$1
IP67 ABS enclosure	Weatherproofing	~\$5
PCB, connectors, passives	Assembly	~\$5
Total BOM		~\$35

3. MARKET ANALYSIS

Total Addressable Market (TAM)

- **Global boat monitoring market:** \$683 million (2025), growing to \$1.72 billion by 2033 at 12.25% CAGR (*Straits Research, 2025*)
- **US boat monitoring market:** \$225 million (2024), projected to reach \$483 million by 2031 at 11.4% CAGR (*Straits Research, 2026*)
- **Connected boat market (broader):** \$7.1 billion globally in 2024, projected \$21.2 billion by 2033 at 13.2% CAGR (*DataIntel, 2024*)

Serviceable Addressable Market (SAM)

- **US registered boats:** 11.8 million (NMMA, 2024)
- **Florida alone:** 1.2 million registered boats (10% of US fleet)
- **New powerboats sold in 2024:** 238,000+ units (NMMA)
- **Fishing boats sold:** 79,269 units (freshwater + saltwater combined)
- **Target SAM:** Powerboat owners at marina slips or regular dock storage ≈ estimated 3-4 million boats

- **SAM value at \$199:** \$597M–\$796M

Serviceable Obtainable Market (SOM) — Realistic 5-Year Target

Year	Units Sold	Revenue	Notes
Year 1 (Kickstarter + post-campaign)	2,500	\$497,500	Campaign + 6-month post-campaign DTC
Year 2	8,000	\$1,592,000	Amazon + DTC + early marina partnerships
Year 3	18,000	\$3,582,000	Retail (West Marine, Bass Pro), influencer reach
Year 4	35,000	\$6,965,000	National retail + dealer channel
Year 5	60,000	\$11,940,000	International expansion + v2 product line

4. COMPETITIVE ANALYSIS

Product	Upfront Cost	Monthly Sub	BYO SIM	Open Protocol	Verdict
BilgeGuard	\$199	\$0 (BYO ~\$2-4)	Yes	Yes	Our product
Siren Marine 3 Pro	\$799	\$18	No	No	4x the price, subscription required
BRNKL (basic)	\$299–\$499	\$20	No	No	Still subscription-locked
Garmin OnDeck	\$349	\$25	No	No	Garmin ecosystem only
DIY (Arduino/RPi)	\$50–\$100	\$2–4	Yes	Yes	Requires technical expertise

Our moat: BilgeGuard is the only product combining (1) sub-\$200 hardware, (2) no mandatory subscription, (3) BYO-SIM flexibility, and (4) plug-and-play installation for non-technical boat owners.

5. BUSINESS MODEL & UNIT ECONOMICS

Revenue Streams

- 1. **Hardware sales** — primary revenue driver (\$199 MSRP)
- 2. **Optional accessories** — additional sensors (temperature, shore power, door alarm): \$19-\$39 each
- 3. **Optional managed SIM** — white-labeled SIM for users who want zero setup: \$3.99/month (future, Year 2+)
- 4. **v2 hardware** — BilgeGuard Pro with camera, relay control, NMEA 2000 integration: \$299 (Year 3+)

Unit Economics (Base Hardware)

Item	Per Unit
Selling price	\$199.00
BOM cost	\$35.00
Assembly + test	\$8.00
Packaging + manual	\$4.00
Fulfillment (ship)	\$9.00
Payment processing (3%)	\$5.97
Gross profit	\$137.03
Gross margin	~69%

Kickstarter fees (5%) reduce first-run margin to ~64% on campaign units — still exceptional for hardware.

Break-Even Analysis

- Fixed costs (Year 1): ~\$45,000 (tooling finalization, certifications, warehouse, basic ops)
- Break-even unit count: ~329 units
- Kickstarter goal of \$50,000 = 251 units — **campaign designed to exceed break-even on Day 1 of shipping**

6. FINANCIAL PROJECTIONS

5-Year P&L Summary (Conservative)

	Year 1	Year 2	Year 3	Year 4	Year 5
Units	2,500	8,000	18,000	35,000	60,000
Revenue	\$497,500	\$1,592,000	\$3,582,000	\$6,965,000	\$11,940,000
COGS	\$154,000	\$448,000	\$972,000	\$1,855,000	\$3,120,000
Gross Profit	\$343,500	\$1,144,000	\$2,610,000	\$5,110,000	\$8,820,000
Operating Expenses	\$180,000	\$420,000	\$780,000	\$1,400,000	\$2,200,000
EBITDA	\$163,500	\$724,000	\$1,830,000	\$3,710,000	\$6,620,000
EBITDA Margin	33%	45%	51%	53%	55%

Key Assumptions

- Average selling price holds at \$199 through Year 3, drops to \$185 in Years 4–5 due to volume discounts to retail
- COGS per unit decreases from \$56 (small batch) to \$52 (Year 5) via volume and supplier negotiation
- OpEx includes marketing (35% of budget), salaries, software/cloud, and customer support
- No external funding assumed — entirely bootstrapped from Kickstarter and reinvested profits

7. FUNDING REQUIREMENTS — KICKSTARTER

Campaign Goal: \$50,000

This is deliberately conservative to ensure fast funding and algorithmic momentum on Kickstarter.

Use of Kickstarter Funds:

Item	Amount
First production run (500 units PCB + assembly)	\$17,500
FCC/IC certification	\$8,000
IP67 enclosure tooling	\$5,000
Fulfillment and shipping (campaign backers)	\$4,500
Campaign video production	\$3,500
Buffer / contingency (10%)	\$5,000
Working capital	\$6,500
Total	\$50,000

Reward Tiers

Tier	Price	Description	Estimated Take
Early Bird (100 units)	\$129	BilgeGuard unit — 35% off	\$12,900
Standard Backer	\$149	BilgeGuard unit — 25% off	primary tier
Dock Pack (2 units)	\$269	Two units for multi-boat owners	upsell
Marina Pack (5 units)	\$599	For marina operators/dealers	B2B seed

8. CERTIFICATIONS & COMPLIANCE ROADMAP

Cert	Purpose	Timeline	Cost
FCC Part 15	Required for cellular radio in US	Month 4-6	~\$5,000-\$8,000
IC (Canada)	Canadian market access	Month 6-8	~\$2,000
CE (Europe)	EU market access (Year 2)	Month 12-18	~\$8,000
RoHS	Required for EU/retail	Month 6	Included in CE
UL Listing	Required for some retail channels	Year 2	~\$10,000

Note: FCC certification is mandatory before retail sales. Kickstarter campaign units can ship under FCC Part 15 self-declaration with proper labeling while formal cert is in process, commonly used by hardware startups.

9. GO-TO-MARKET STRATEGY

Phase 1: Kickstarter (Months 1-5)

- Pre-launch email list building via \$5 VIP reservation model (Blazon Agency methodology)
- YouTube and Reddit seeding in boating communities (r/boating, r/liveaboard, The Hull Truth)
- Target: 500 email reservations before launch day
- Campaign video: emotional hook (real story of a boat sinking at the dock) + product demo
- Launch with Early Bird pricing to hit 100% in 48 hours

Phase 2: DTC + Amazon (Months 6-18)

- Launch BilgeGuard.com Shopify store post-campaign
- Amazon FBA listing with optimized keywords ("boat bilge alarm," "boat GPS tracker no subscription")
- YouTube channel: installation walkthroughs, comparison videos
- Target boating influencers (Salt Water Experience, Wired2Fish) for product reviews

Phase 3: Retail + B2B (Months 18-36)

- West Marine buyer pitch (they carry ~\$200 electronics consistently)
- Bass Pro Shops marine electronics section
- Marina operator bulk pricing (5-10 unit marina packs for slip renters)
- Boat dealer OEM bundle discussions (similar to how Garmin is bundled with boats)

Phase 4: International + v2 (Years 3-5)

- Canada launch (similar fleet demographics, bilingual packaging)
- Australia/NZ (strong boating culture, English market)
- BilgeGuard Pro: adds NMEA 2000 integration, relay control, camera support

10. RISKS & MITIGATION

Risk	Likelihood	Impact	Mitigation
FCC certification delay	Medium	High	Begin cert process in Month 1, pre-brief lab, budget contingency
LTE-M network coverage gaps	Low	Medium	Support 4G LTE fallback; WiFi mode covers marinas
Component supply chain disruption	Medium	Medium	6-month BOM inventory buffer after campaign
Competitor price cut	Medium	Medium	Open-source firmware builds community moat
Counterfeit clones (China)	Medium	Low	Brand trust, community, certifications as differentiators
Kickstarter campaign underperforms	Low	High	\$5 VIP list strategy, pre-built audience before launch

11. FOUNDING TEAM REQUIREMENTS

For a solo or small-team launch, required competencies:

- **Hardware engineer** — ESP32/PCB design, LTE-M integration (*can be contracted*)
- **Firmware engineer** — MQTT, cellular stack, OTA updates (*can be contracted*)
- **Founder** — product vision, marketing, community, campaign management
- **Optional:** Marketing specialist for Kickstarter campaign video and pre-launch

SOURCES

- NMMA 2024 U.S. Recreational Boating Statistical Abstract
- Straits Research: Boat Monitoring Market Report (2025–2026)
- Spherical Insights: Global Boat Monitoring Market 2026–2035
- DataIntel: Connected Boat Market Research Report 2033
- US Coast Guard Recreational Boating Statistics 2024
- Salt Water Sportsman: Boat Security System Review (2023)
- Blazon Agency: How to Crowdfund a Hardware Product in 2026
- CrowdCrux: Kickstarter Stats 2024
- Community data: The Hull Truth, Trawler Forum, Reddit r/boating